Question 1

Display running processes and identify the first and last process in the list

Approach

I used the `ps` command with the `aux` option to display all active processes running on the system and observed which process appears first and which appears last.

Commands

- `ps` → shows active processes
- `aux` → displays all processes for all users with detailed information

Output

STEP 1 : Display all running processes

```
┌──(aman㉿kali)-[~]
└─$ ps aux
USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1  0.0  0.1  24556 15856 ?        Ss   10:16   0:03 /sbin/init splash
root           2  0.0  0.0      0     0 ?        S    10:16   0:00 [kthreadd]
root           3  0.0  0.0      0     0 ?        S    10:16   0:00 [pool_workqueue_release]
root           4  0.0  0.0      0     0 ?        I<   10:16   0:00 [kworker/R-rcu_gp]
root           5  0.0  0.0      0     0 ?        I<   10:16   0:00 [kworker/R-sync_wq]
root           6  0.0  0.0      0     0 ?        I<   10:16   0:00 [kworker/R-kvfree_rcu_reclaim]
root           7  0.0  0.0      0     0 ?        I<   10:16   0:00 [kworker/R-slub_flushwq]
root           8  0.0  0.0      0     0 ?        I<   10:16   0:00 [kworker/R-netns]
root          10  0.0  0.0      0     0 ?        I<   10:16   0:00 [kworker/0:0H-kblockd]
root          13  0.0  0.0      0     0 ?        I<   10:16   0:00 [kworker/R-mm_percpu_wq]
root          14  0.0  0.0      0     0 ?        S    10:16   0:00 [ksoftirqd/0]
root          15  0.0  0.0      0     0 ?        R    10:16   0:00 [rcu_preempt]
root          16  0.0  0.0      0     0 ?        S    10:16   0:00 [rcu_exp_par_gp_kthread_worker/0]
root          17  0.0  0.0      0     0 ?        S    10:16   0:00 [rcu_exp_gp_kthread_worker]
root          18  0.0  0.0      0     0 ?        S    10:16   0:00 [migration/0]
root          19  0.0  0.0      0     0 ?        S    10:16   0:00 [idle_inject/0]
root          20  0.0  0.0      0     0 ?        S    10:16   0:00 [cpuhp/0]
root          22  0.0  0.0      0     0 ?        S    10:16   0:00 [kdevtmpfs]
root          23  0.0  0.0      0     0 ?        I<   10:16   0:00 [kworker/R-inet_frag_wq]
root          24  0.0  0.0      0     0 ?        I    10:16   0:00 [rcu_tasks_kthread]
root          25  0.0  0.0      0     0 ?        I    10:16   0:00 [rcu_tasks_rude_kthread]
root          26  0.0  0.0      0     0 ?        I    10:16   0:00 [rcu_tasks_trace_kthread]
root          27  0.0  0.0      0     0 ?        S    10:16   0:00 [kauditd]
root          28  0.0  0.0      0     0 ?        S    10:16   0:00 [khungtaskd]
root          29  0.0  0.0      0     0 ?        S    10:16   0:00 [oom_reaper]
root          31  0.0  0.0      0     0 ?        I    10:16   0:00 [kworker/u4:2-events_unbound]
root          32  0.0  0.0      0     0 ?        I<   10:16   0:00 [kworker/R-writeback]
root          33  0.0  0.0      0     0 ?        S    10:16   0:01 [kcompactd0]
root          34  0.0  0.0      0     0 ?        SN   10:16   0:00 [ksmd]
root          35  0.0  0.0      0     0 ?        SN   10:16   0:00 [khugepaged]
root          36  0.0  0.0      0     0 ?        I<   10:16   0:00 [kworker/R-kblockd]
root          37  0.0  0.0      0     0 ?        I<   10:16   0:00 [kworker/R-blkcg_punt_bio]
root          38  0.0  0.0      0     0 ?        I<   10:16   0:00 [kworker/R-kintegrityd]
root          39  0.0  0.0      0     0 ?        S    10:16   0:00 [irq/9-acpi]
root          40  0.0  0.0      0     0 ?        I<   10:16   0:00 [kworker/R-tpm_dev_wq]
root          41  0.0  0.0      0     0 ?        I<   10:16   0:00 [kworker/R-edac-poller]
root          42  0.0  0.0      0     0 ?        I<   10:16   0:00 [kworker/R-devfreq_wq]
root          43  0.0  0.0      0     0 ?        I<   10:16   0:00 [kworker/R-quota_events_unbound]
root          45  0.0  0.0      0     0 ?        S    10:16   0:00 [kswapd0]
root          46  0.0  0.0      0     0 ?        I    10:16   0:00 [kdamond.0]
root          49  0.0  0.0      0     0 ?        I<   10:16   0:00 [kworker/R-kthrotld]
root          50  0.0  0.0      0     0 ?        I<   10:16   0:00 [kworker/R-acpi_thermal_pm]
root          51  0.0  0.0      0     0 ?        I<   10:16   0:00 [kworker/R-mld]
root          52  0.0  0.0      0     0 ?        I<   10:16   0:00 [kworker/R-ipv6_addrconf]
root          53  0.0  0.0      0     0 ?        I<   10:16   0:00 [kworker/R-kstrp]
root          56  0.0  0.0      0     0 ?        I    10:16   0:00 [kworker/u4:3-flush-8:0]
root          61  0.0  0.0      0     0 ?        I<   10:16   0:00 [kworker/u5:0-ttm]
root         193  0.0  0.0      0     0 ?        I<   10:16   0:00 [kworker/R-ata_sff]
root         194  0.0  0.0      0     0 ?        S    10:16   0:00 [scsi_eh_0]
root         195  0.0  0.0      0     0 ?        I<   10:16   0:00 [kworker/R-scsi_tmf_0]
```

STEP 2 : Identify first and last process

```
┌──(aman㉿kali)-[~]
└─$ ps -p 1 -o pid,ppid,cmd
    PID    PPID CMD
      1       0 /sbin/init splash
```

```
┌──(aman㉿kali)-[~]
└─$ ps -eo pid,ppid,cmd --sort=pid | tail -1
  15639    3609 tail -1
```

Key Observations

Processes are displayed with :

- PID → process ID
- CPU usage
- memory usage
- command name

What I learned

Linux allows monitoring of all running processes including system and user processes.

Question 2

Identify the two processes consuming the most system resources

Approach

I used the `top` command to monitor real-time system resource usage and identified the two processes using the highest CPU or memory.

Commands

- `top` → shows live system process activity

Output

STEP 1 : open process monitor

```
┌──(aman㉿kali)-[~]
└─$ top
top - 17:10:36 up  6:54,  1 user,  load average: 0.02, 0.01, 0.00
Tasks: 166 total,   1 running, 165 sleeping,   0 stopped,   0 zombie
%Cpu(s):  1.4 us,  1.4 sy,  0.0 ni, 97.3 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :   7946.2 total,   6391.8 free,    943.4 used,    867.1 buff/cache
MiB Swap:   1654.0 total,   1654.0 free,      0.0 used.   7002.8 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
    696 root      20   0  483692 177720  86356 S   1.0   2.2   0:21.86 Xorg
    603 root      20   0  352936   2948   2684 S   0.3   0.0   0:14.06 VBoxDRMClient
   2218 aman      20   0  215380   3296   2884 S   0.3   0.0   0:03.39 VBoxClient
   2402 aman      20   0  293580  42208  22160 S   0.3   0.5   0:03.99 wrapper-2.0
   2404 aman      20   0  277080  30480  23172 S   0.3   0.4   0:03.56 wrapper-2.0
   3553 aman      20   0  593420  72760  55680 S   0.3   0.9   0:02.93 qterminal
      1 root      20   0   25196  16324  11472 S   0.0   0.2   0:05.09 systemd
      2 root      20   0       0      0      0 S   0.0   0.0   0:00.00 kthreadd
      3 root      20   0       0      0      0 S   0.0   0.0   0:00.00 pool_workqueue_release
      4 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-rcu_gp
      5 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-sync_wq
      6 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-kvfree_rcu_reclaim
      7 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-slub_flushwq
      8 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-netns
     10 root       0 -20       0      0      0 I   0.0   0.0   0:00.15 kworker/0:0H-kblockd
     13 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-mm_percpu_wq
     14 root      20   0       0      0      0 S   0.0   0.0   0:00.26 ksoftirqd/0
     15 root      20   0       0      0      0 I   0.0   0.0   0:00.83 rcu_preempt
     16 root      20   0       0      0      0 S   0.0   0.0   0:00.00 rcu_exp_par_gp_kthread_worker/0
     17 root      20   0       0      0      0 S   0.0   0.0   0:00.00 rcu_exp_gp_kthread_worker
     18 root      rt   0       0      0      0 S   0.0   0.0   0:00.14 migration/0
     19 root     -51   0       0      0      0 S   0.0   0.0   0:00.00 idle_inject/0
     20 root      20   0       0      0      0 S   0.0   0.0   0:00.00 cpuhp/0
     22 root      20   0       0      0      0 S   0.0   0.0   0:00.00 kdevtmpfs
     23 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-inet_frag_wq
     24 root      20   0       0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_kthread
     25 root      20   0       0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_rude_kthread
     26 root      20   0       0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_trace_kthread
     27 root      20   0       0      0      0 S   0.0   0.0   0:00.00 kauditd
     28 root      20   0       0      0      0 S   0.0   0.0   0:00.01 khungtaskd
     29 root      20   0       0      0      0 S   0.0   0.0   0:00.00 oom_reaper
     31 root      20   0       0      0      0 I   0.0   0.0   0:00.50 kworker/u4:2-events_unbound
     32 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-writeback
     33 root      20   0       0      0      0 S   0.0   0.0   0:01.55 kcompactd0
     34 root      25   5       0      0      0 S   0.0   0.0   0:00.00 ksmd
     35 root      39  19       0      0      0 S   0.0   0.0   0:00.85 khugepaged
     36 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-kblockd
     37 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-blkcg_punt_bio
     38 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-kintegrityd
     39 root     -51   0       0      0      0 S   0.0   0.0   0:00.00 irq/9-acpi
     40 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-tpm_dev_wq
     41 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-edac-poller
     42 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-devfreq_wq
```

STEP 2 : identify highest resource processes

```
  PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
  696 root      20   0  483692 177720  86356 S   1.3   2.2   0:24.15 Xorg
```

Key Observations

Processes are dynamically sorted based on CPU usage.

What I learned

Some processes consume more resource and may affect system performance.

Question 3

Terminate the process consuming the most resources

Approach

I identified the process ID (PID) of the resource-heavy process and terminated it using the kill command.

Commands

- `kill` → stops a process
- `-9` → force stop process

Output

STEP 1 : find PID using top

```
┌──(aman㉿kali)-[~]
└─$ top
top - 17:10:36 up  6:54,  1 user,  load average: 0.02, 0.01, 0.00
Tasks: 166 total,   1 running, 165 sleeping,   0 stopped,   0 zombie
%Cpu(s):  1.4 us,  1.4 sy,  0.0 ni, 97.3 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :   7946.2 total,   6391.8 free,    943.4 used,    867.1 buff/cache
MiB Swap:   1654.0 total,   1654.0 free,      0.0 used.   7002.8 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
    696 root      20   0  483692 177720  86356 S   1.0   2.2   0:21.86 Xorg
    603 root      20   0  352936   2948   2684 S   0.3   0.0   0:14.06 VBoxDRMClient
   2218 aman      20   0  215380   3296   2884 S   0.3   0.0   0:03.39 VBoxClient
   2402 aman      20   0  293580  42208  22160 S   0.3   0.5   0:03.99 wrapper-2.0
   2404 aman      20   0  277080  30480  23172 S   0.3   0.4   0:03.56 wrapper-2.0
   3553 aman      20   0  593420  72760  55680 S   0.3   0.9   0:02.93 qterminal
      1 root      20   0   25196  16324  11472 S   0.0   0.2   0:05.09 systemd
      2 root      20   0       0      0      0 S   0.0   0.0   0:00.00 kthreadd
      3 root      20   0       0      0      0 S   0.0   0.0   0:00.00 pool_workqueue_release
      4 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-rcu_gp
      5 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-sync_wq
      6 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-kvfree_rcu_reclaim
      7 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-slub_flushwq
      8 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-netns
     10 root       0 -20       0      0      0 I   0.0   0.0   0:00.15 kworker/0:0H-kblockd
     13 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-mm_percpu_wq
     14 root      20   0       0      0      0 S   0.0   0.0   0:00.26 ksoftirqd/0
     15 root      20   0       0      0      0 I   0.0   0.0   0:00.83 rcu_preempt
     16 root      20   0       0      0      0 S   0.0   0.0   0:00.00 rcu_exp_par_gp_kthread_worker/0
     17 root      20   0       0      0      0 S   0.0   0.0   0:00.00 rcu_exp_gp_kthread_worker
     18 root      rt   0       0      0      0 S   0.0   0.0   0:00.14 migration/0
     19 root     -51   0       0      0      0 S   0.0   0.0   0:00.00 idle_inject/0
     20 root      20   0       0      0      0 S   0.0   0.0   0:00.00 cpuhp/0
     22 root      20   0       0      0      0 S   0.0   0.0   0:00.00 kdevtmpfs
     23 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-inet_frag_wq
     24 root      20   0       0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_kthread
     25 root      20   0       0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_rude_kthread
     26 root      20   0       0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_trace_kthread
     27 root      20   0       0      0      0 S   0.0   0.0   0:00.00 kauditd
     28 root      20   0       0      0      0 S   0.0   0.0   0:00.01 khungtaskd
     29 root      20   0       0      0      0 S   0.0   0.0   0:00.00 oom_reaper
     31 root      20   0       0      0      0 I   0.0   0.0   0:00.50 kworker/u4:2-events_unbound
     32 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-writeback
     33 root      20   0       0      0      0 S   0.0   0.0   0:01.55 kcompactd0
     34 root      25   5       0      0      0 S   0.0   0.0   0:00.00 ksmd
     35 root      39  19       0      0      0 S   0.0   0.0   0:00.85 khugepaged
     36 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-kblockd
     37 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-blkcg_punt_bio
     38 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-kintegrityd
     39 root     -51   0       0      0      0 S   0.0   0.0   0:00.00 irq/9-acpi
     40 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-tpm_dev_wq
     41 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-edac-poller
     42 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-devfreq_wq
```

STEP 2 : terminate process

```
┌──(aman㉿kali)-[~]
└─$ kill -9 2402
```

STEP 3 : confirm process stopped

```
┌──(aman㉿kali)-[~]
└─$ top
top - 17:27:48 up  7:11,  1 user,  load average: 0.80, 0.28, 0.14
Tasks: 167 total,   1 running, 166 sleeping,   0 stopped,   0 zombie
%Cpu(s):  4.7 us,  3.7 sy,  0.0 ni, 91.6 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :   7946.2 total,   6377.8 free,    954.1 used,    870.9 buff/cache
MiB Swap:   1654.0 total,   1654.0 free,      0.0 used.   6992.1 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
    696 root      20   0  484204 177744  86368 S   4.3   2.2   0:37.06 Xorg
   3553 aman      20   0  593008  72760  55652 S   2.0   0.9   0:06.51 qterminal
  23441 aman      20   0  293584  42212  22080 S   0.7   0.5   0:01.09 wrapper-2.0
   2404 aman      20   0  277080  30480  23172 S   0.3   0.4   0:05.80 wrapper-2.0
      1 root      20   0   25196  16324  11472 S   0.0   0.2   0:05.12 systemd
      2 root      20   0       0      0      0 S   0.0   0.0   0:00.00 kthreadd
      3 root      20   0       0      0      0 S   0.0   0.0   0:00.00 pool_workqueue_release
      4 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-rcu_gp
      5 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-sync_wq
      6 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-kvfree_rcu_reclaim
      7 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-slub_flushwq
      8 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-netns
     10 root       0 -20       0      0      0 I   0.0   0.0   0:00.17 kworker/0:0H-kblockd
     13 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-mm_percpu_wq
     14 root      20   0       0      0      0 S   0.0   0.0   0:00.39 ksoftirqd/0
     15 root      20   0       0      0      0 I   0.0   0.0   0:01.01 rcu_preempt
     16 root      20   0       0      0      0 S   0.0   0.0   0:00.00 rcu_exp_par_gp_kthread_worker/0
     17 root      20   0       0      0      0 S   0.0   0.0   0:00.00 rcu_exp_gp_kthread_worker
     18 root      rt   0       0      0      0 S   0.0   0.0   0:00.15 migration/0
     19 root     -51   0       0      0      0 S   0.0   0.0   0:00.00 idle_inject/0
     20 root      20   0       0      0      0 S   0.0   0.0   0:00.00 cpuhp/0
     22 root      20   0       0      0      0 S   0.0   0.0   0:00.00 kdevtmpfs
     23 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-inet_frag_wq
     24 root      20   0       0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_kthread
     25 root      20   0       0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_rude_kthread
     26 root      20   0       0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_trace_kthread
     27 root      20   0       0      0      0 S   0.0   0.0   0:00.00 kauditd
     28 root      20   0       0      0      0 S   0.0   0.0   0:00.01 khungtaskd
     29 root      20   0       0      0      0 S   0.0   0.0   0:00.00 oom_reaper
     31 root      20   0       0      0      0 I   0.0   0.0   0:00.52 kworker/u4:2-kvfree_rcu_reclaim
     32 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-writeback
     33 root      20   0       0      0      0 S   0.0   0.0   0:01.59 kcompactd0
     34 root      25   5       0      0      0 S   0.0   0.0   0:00.00 ksmd
     35 root      39  19       0      0      0 S   0.0   0.0   0:00.87 khugepaged
```

Key Observations

Process disappears from the process list after termination.

What I learned

Processes can be manually stopped to free system resources.

Question 4

Reduce the priority of a running process

Approach

I used the renice command to change the priority level of a running process to 19 (lowest priority).

Commands

- `renice` → modifies process priority
- nice values range :
  - -20 → highest priority
  - 19 → lowest priority

Output

STEP 1 : find PID

```
┌──(aman㉿kali)-[~]
└─$ ps aux
USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1  0.0  0.2  25196 16324 ?        Ss   10:16   0:05 /sbin/init splash
root           2  0.0  0.0      0     0 ?        S    10:16   0:00 [kthreadd]
root           3  0.0  0.0      0     0 ?        S    10:16   0:00 [pool_workqueue_release]
root           4  0.0  0.0      0     0 ?        I<   10:16   0:00 [kworker/R-rcu_gp]
root           5  0.0  0.0      0     0 ?        I<   10:16   0:00 [kworker/R-sync_wq]
root           6  0.0  0.0      0     0 ?        I<   10:16   0:00 [kworker/R-kvfree_rcu_reclaim]
root           7  0.0  0.0      0     0 ?        I<   10:16   0:00 [kworker/R-slub_flushwq]
root           8  0.0  0.0      0     0 ?        I<   10:16   0:00 [kworker/R-netns]
root          10  0.0  0.0      0     0 ?        I<   10:16   0:00 [kworker/0:0H-kblockd]
root          13  0.0  0.0      0     0 ?        I<   10:16   0:00 [kworker/R-mm_percpu_wq]
root          14  0.0  0.0      0     0 ?        S    10:16   0:00 [ksoftirqd/0]
root          15  0.0  0.0      0     0 ?        I    10:16   0:01 [rcu_preempt]
root          16  0.0  0.0      0     0 ?        S    10:16   0:00 [rcu_exp_par_gp_kthread_worker/0]
root          17  0.0  0.0      0     0 ?        S    10:16   0:00 [rcu_exp_gp_kthread_worker]
root          18  0.0  0.0      0     0 ?        S    10:16   0:00 [migration/0]
root          19  0.0  0.0      0     0 ?        S    10:16   0:00 [idle_inject/0]
root          20  0.0  0.0      0     0 ?        S    10:16   0:00 [cpuhp/0]
root          22  0.0  0.0      0     0 ?        S    10:16   0:00 [kdevtmpfs]
root          23  0.0  0.0      0     0 ?        I<   10:16   0:00 [kworker/R-inet_frag_wq]
root          24  0.0  0.0      0     0 ?        I    10:16   0:00 [rcu_tasks_kthread]
root          25  0.0  0.0      0     0 ?        I    10:16   0:00 [rcu_tasks_rude_kthread]
root          26  0.0  0.0      0     0 ?        I    10:16   0:00 [rcu_tasks_trace_kthread]
root          27  0.0  0.0      0     0 ?        S    10:16   0:00 [kauditd]
root          28  0.0  0.0      0     0 ?        S    10:16   0:00 [khungtaskd]
root          29  0.0  0.0      0     0 ?        S    10:16   0:00 [oom_reaper]
root          31  0.0  0.0      0     0 ?        I    10:16   0:00 [kworker/u4:2-events_unbound]
root          32  0.0  0.0      0     0 ?        I<   10:16   0:00 [kworker/R-writeback]
root          33  0.0  0.0      0     0 ?        S    10:16   0:01 [kcompactd0]
root          34  0.0  0.0      0     0 ?        SN   10:16   0:00 [ksmd]
root          35  0.0  0.0      0     0 ?        SN   10:16   0:00 [khugepaged]
root          36  0.0  0.0      0     0 ?        I<   10:16   0:00 [kworker/R-kblockd]
root          37  0.0  0.0      0     0 ?        I<   10:16   0:00 [kworker/R-blkcg_punt_bio]
root          38  0.0  0.0      0     0 ?        I<   10:16   0:00 [kworker/R-kintegrityd]
root          39  0.0  0.0      0     0 ?        S    10:16   0:00 [irq/9-acpi]
root          40  0.0  0.0      0     0 ?        I<   10:16   0:00 [kworker/R-tpm_dev_wq]
root          41  0.0  0.0      0     0 ?        I<   10:16   0:00 [kworker/R-edac-poller]
root          42  0.0  0.0      0     0 ?        I<   10:16   0:00 [kworker/R-devfreq_wq]
root          43  0.0  0.0      0     0 ?        I<   10:16   0:00 [kworker/R-quota_events_unbound]
root          45  0.0  0.0      0     0 ?        S    10:16   0:00 [kswapd0]
root          46  0.0  0.0      0     0 ?        I    10:16   0:00 [kdamond.0]
root          49  0.0  0.0      0     0 ?        I<   10:16   0:00 [kworker/R-kthrotld]
root          50  0.0  0.0      0     0 ?        I<   10:16   0:00 [kworker/R-acpi_thermal_pm]
root          51  0.0  0.0      0     0 ?        I<   10:16   0:00 [kworker/R-mld]
root          52  0.0  0.0      0     0 ?        I<   10:16   0:00 [kworker/R-ipv6_addrconf]
root          53  0.0  0.0      0     0 ?        I<   10:16   0:00 [kworker/R-kstrp]
root          56  0.0  0.0      0     0 ?        I    10:16   0:00 [kworker/u4:3-kvfree_rcu_reclaim]
root          61  0.0  0.0      0     0 ?        I<   10:16   0:00 [kworker/u5:0-ttm]
```

STEP 2 : change process priority

```
┌──(aman㉿kali)-[~]
└─$ renice 19 2605
2605 (process ID) old priority 0, new priority 19
```

STEP 3 : Verify change

```
┌──(aman㉿kali)-[~]
└─$ ps aux
USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1  0.0  0.2  25196 16324 ?        Ss   10:16   0:05 /sbin/init splash
root           2  0.0  0.0      0     0 ?        S    10:16   0:00 [kthreadd]
root           3  0.0  0.0      0     0 ?        S    10:16   0:00 [pool_workqueue_release]
root           4  0.0  0.0      0     0 ?        I<   10:16   0:00 [kworker/R-rcu_gp]
root           5  0.0  0.0      0     0 ?        I<   10:16   0:00 [kworker/R-sync_wq]
root           6  0.0  0.0      0     0 ?        I<   10:16   0:00 [kworker/R-kvfree_rcu_reclaim]
root           7  0.0  0.0      0     0 ?        I<   10:16   0:00 [kworker/R-slub_flushwq]
root           8  0.0  0.0      0     0 ?        I<   10:16   0:00 [kworker/R-netns]
root          10  0.0  0.0      0     0 ?        I<   10:16   0:00 [kworker/0:0H-kblockd]
root          13  0.0  0.0      0     0 ?        I<   10:16   0:00 [kworker/R-mm_percpu_wq]
root          14  0.0  0.0      0     0 ?        S    10:16   0:00 [ksoftirqd/0]
root          15  0.0  0.0      0     0 ?        I    10:16   0:01 [rcu_preempt]
root          16  0.0  0.0      0     0 ?        S    10:16   0:00 [rcu_exp_par_gp_kthread_worker/0]
root          17  0.0  0.0      0     0 ?        S    10:16   0:00 [rcu_exp_gp_kthread_worker]
root          18  0.0  0.0      0     0 ?        S    10:16   0:00 [migration/0]
root          19  0.0  0.0      0     0 ?        S    10:16   0:00 [idle_inject/0]
root          20  0.0  0.0      0     0 ?        S    10:16   0:00 [cpuhp/0]
root          22  0.0  0.0      0     0 ?        S    10:16   0:00 [kdevtmpfs]
root          23  0.0  0.0      0     0 ?        I<   10:16   0:00 [kworker/R-inet_frag_wq]
root          24  0.0  0.0      0     0 ?        I    10:16   0:00 [rcu_tasks_kthread]
root          25  0.0  0.0      0     0 ?        I    10:16   0:00 [rcu_tasks_rude_kthread]
root          26  0.0  0.0      0     0 ?        I    10:16   0:00 [rcu_tasks_trace_kthread]
root          27  0.0  0.0      0     0 ?        S    10:16   0:00 [kauditd]
root          28  0.0  0.0      0     0 ?        S    10:16   0:00 [khungtaskd]
root          29  0.0  0.0      0     0 ?        S    10:16   0:00 [oom_reaper]
root          31  0.0  0.0      0     0 ?        I    10:16   0:00 [kworker/u4:2-flush-8:0]
root          32  0.0  0.0      0     0 ?        I<   10:16   0:00 [kworker/R-writeback]
root          33  0.0  0.0      0     0 ?        S    10:16   0:01 [kcompactd0]
root          34  0.0  0.0      0     0 ?        SN   10:16   0:00 [ksmd]
root          35  0.0  0.0      0     0 ?        SN   10:16   0:00 [khugepaged]
root          36  0.0  0.0      0     0 ?        I<   10:16   0:00 [kworker/R-kblockd]
root          37  0.0  0.0      0     0 ?        I<   10:16   0:00 [kworker/R-blkcg_punt_bio]
root          38  0.0  0.0      0     0 ?        I<   10:16   0:00 [kworker/R-kintegrityd]
root          39  0.0  0.0      0     0 ?        S    10:16   0:00 [irq/9-acpi]
root          40  0.0  0.0      0     0 ?        I<   10:16   0:00 [kworker/R-tpm_dev_wq]
root          41  0.0  0.0      0     0 ?        I<   10:16   0:00 [kworker/R-edac-poller]
root          42  0.0  0.0      0     0 ?        I<   10:16   0:00 [kworker/R-devfreq_wq]
root          43  0.0  0.0      0     0 ?        I<   10:16   0:00 [kworker/R-quota_events_unbound]
root          45  0.0  0.0      0     0 ?        S    10:16   0:00 [kswapd0]
root          46  0.0  0.0      0     0 ?        I    10:16   0:00 [kdamond.0]
root          49  0.0  0.0      0     0 ?        I<   10:16   0:00 [kworker/R-kthrotld]
root          50  0.0  0.0      0     0 ?        I<   10:16   0:00 [kworker/R-acpi_thermal_pm]
root          51  0.0  0.0      0     0 ?        I<   10:16   0:00 [kworker/R-mld]
root          52  0.0  0.0      0     0 ?        I<   10:16   0:00 [kworker/R-ipv6_addrconf]
root          53  0.0  0.0      0     0 ?        I<   10:16   0:00 [kworker/R-kstrp]
root          56  0.0  0.0      0     0 ?        I    10:16   0:00 [kworker/u4:3-kvfree_rcu_reclaim]
root          61  0.0  0.0      0     0 ?        I<   10:16   0:00 [kworker/u5:0-ttm]
root         193  0.0  0.0      0     0 ?        I<   10:16   0:00 [kworker/R-ata_sff]
root         194  0.0  0.0      0     0 ?        S    10:16   0:00 [scsi_eh_0]
```

Key Observation

Lower priority processes receive fewer CPU resources.

What I learned

Process priority can be controlled to optimize system performance.

Question 5

Create a script and schedule it to run automatically at a specific time

Approach

I created a simple bash script and scheduled it using the at command to execute at a specific time.

Commands

- `nano` → create scrip
- `chmod` → make script executable
- `at` → schedule task
- `atq` → view scheduled jobs

Output

STEP 1 : create script file

```
┌──(aman㉿kali)-[~]
└─$ nano myscanning.sh
```

STEP 2 : make script executable

```
┌──(aman㉿kali)-[~]
└─$ chmod +x myscanning.sh
```

STEP 3 : schedule script

```
┌──(aman㉿kali)-[~]
└─$ at 1:00 AM Wed
warning: commands will be executed using /bin/sh
at Wed Apr  1 01:00:00 2026
at> ./myscanning.sh<EOT>
job 1 at Wed Apr  1 01:00:00 2026
```

STEP 4 : verify scheduled job

```
┌──(aman㉿kali)-[~]
└─$ atq
1       Wed Apr  1 01:00:00 2026 a aman
```

Key Observations

Scheduled tasks run automatically without manual execution

What I learned

Task scheduling helps automate system operations and repetitive activities

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Security Insights

- Monitoring running processes helps identify suspicious or malicious programs.
- High resource usage may indicate malware or unauthorized activity.
- Attackers often terminate security tools to avoid detection.
- Changing process priority can be used to weaken security software performance.
- Scheduled scripts can automate attacks or security monitoring tasks.